Chapter 6: t-Tests

Statistics for Environmental Science

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Table of contents

Where Student's *t* came from **2**

From *z* to *t* **2**
- Why t has heavier tails than z . . . 3
- Degrees of freedom . . . 3
- Critical values . . . 4
- t when H_0 is true . . . 4

The one-sample *t*-test **5**
- Calculating t from data . . . 5
- Interpreting t . . . 6
- Getting p-values for t . . . 8
- One-tailed vs. two-tailed, applied . . . 8
- Assumptions, and what to do when they don't hold . . . 9

The paired *t*-test **10**
- Calculating t from the differences . . . 11
- Scaling to the full sample . . . 11

The independent-samples *t*-test **12**

Simulating data for t-tests **15**
- Simulating a one-sample t-test . . . 15
- Simulating paired and independent-samples tests . . . 16

Chapter Summary **16**

Where Student's *t* came from

In the early 1900s, William Sealy Gosset worked as a brewer at Guinness, where quality control depended on testing ingredients such as barley and hops. Every test used up beer or ingredients, so Guinness could only test small samples. The statistics of the time relied on the normal (z) distribution, which works when samples are large.

Gosset set out to measure how much extra uncertainty small samples add, and derived a new sampling distribution for it, now known as the **Student's t distribution**. Guinness restricted outside publication, so he published the 1908 paper under the pseudonym "Student" (Student 1908), which is why we still say Student's t-test. This chapter uses Gosset's approach. We measure a signal, divide it by its standard error, and compare the resulting t to the t distribution for $n - 1$ degrees of freedom.

We often compare a sample mean to a reference (drinking-water standard, historical average) or compare two conditions (before/after, site A vs site B). A t-test asks whether the observed difference is larger than you would expect from sampling variation alone, given your n and variability. Chapter 5 added two questions to every test: how large the effect is, and whether the design had the power to detect it. Both carry over to the t-tests here.

Three common t-tests you will use.

1. **One-sample t-test.** Compare a sample mean to a target (μ_0).
2. **Paired t-test.** Compare before/after (or matched) measurements on the same units by running a one-sample t-test on the *differences*.
3. **Independent-samples t-test (also called two-sample t-test).** Compare means from two groups measured on different units. The pooled version assumes the two groups have similar spread; Welch's version (Welch 1947) does not, and it is what R's `t.test()` runs unless you set `var.equal = TRUE`.

From *z* to *t*

A test statistic is the effect you observed divided by an estimate of how much it could vary by chance:

$$\text{statistic} = \frac{\text{effect}}{\text{error}}$$

In Chapters 4 and 5 we used this with z to test a sample mean against μ_0:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

The numerator is the effect, the distance between the sample mean and the value H_0 names. The denominator is the standard error, the standard deviation of the sampling distribution of the sample mean. The thing being tested is a sample mean, so the yardstick is how far sample means of size n fall from μ_0 by chance. Means vary less than individual observations do, by a factor of $\sqrt{n}$, which is why the denominator is $\sigma/\sqrt{n}$ rather than σ.

z needs the population standard deviation σ, and real studies almost never know it. The t-test estimates it with the sample standard deviation s:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

t still measures the distance from μ_0 in standard errors. The difference is that s is itself an estimate that changes from sample to sample, and the rest of this section shows what that does to the distribution of t.

Why *t* has heavier tails than *z*

Because s varies from sample to sample, the t distribution has heavier tails than the standard normal. As n grows, s stabilizes and t converges to z.

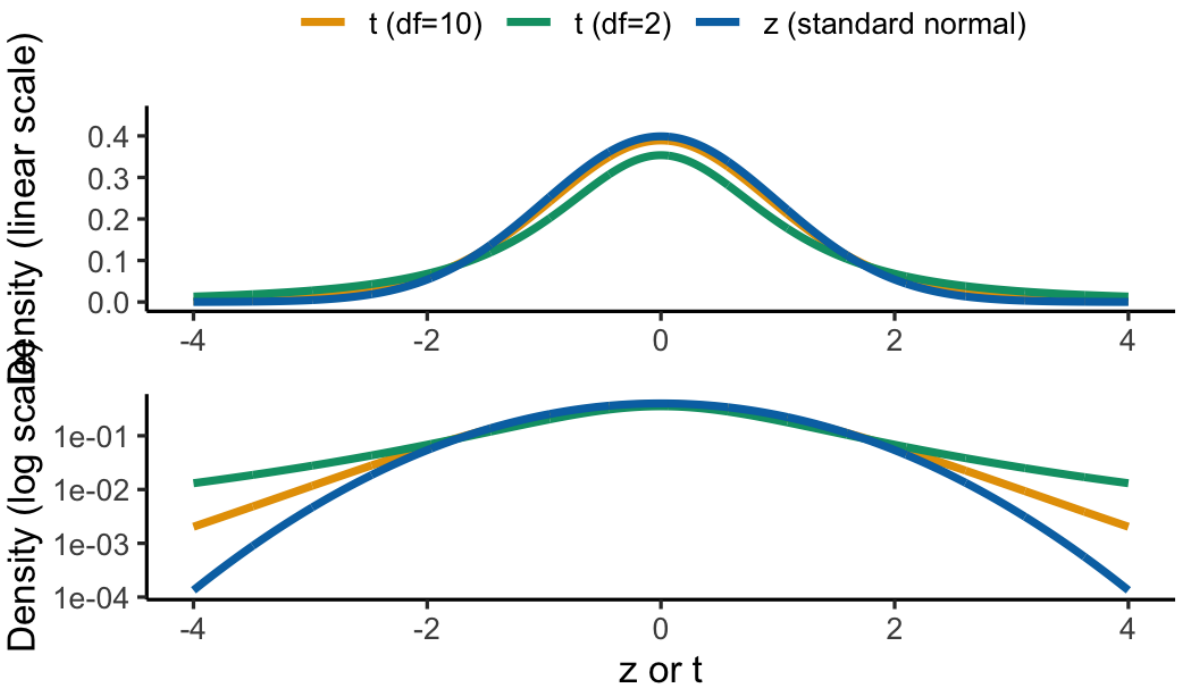


Figure 1: Normal (z) vs. t distributions: on a linear scale (top), tail differences look small; on a log scale (bottom), the heavy tails of small-n t are clear.

> Tip: When is z appropriate?
>
> If the population SD σ is known *and* the sampling distribution of the sample mean is (approximately) normal, use z. In practice, σ is rarely known, so we use t with s and $df = n - 1$.

Degrees of freedom

Figure 1 drew a different t curve for each value of a number called the **degrees of freedom**, df. It is also the number in parentheses whenever a t-test is reported, as in $t(9) = 1.94$.

For one-sample and paired t-tests, $df = n - 1$. A sample of 10 has $df = 9$.

Degrees of freedom counts how many values are still free to vary once something has been estimated from the data. Consider three numbers with a mean of 2: pick the first two freely, say 51 and -3, and the third number is no longer free. It has to be -42 for the mean to come out to 2. Two of the three numbers had freedom; the third didn't. That's $df = n - 1 = 2$ for three numbers.

In a t-test, the mean is the thing estimated, and the df tells you how much information is left for estimating the spread, s. With few degrees of freedom, s can be well above or below σ, so large t values turn up more

often and the t curve has heavier tails. As df grows, s settles close to σ and the t curve approaches the normal. That is why each df gets its own curve.

In a simulation with H_0 true, 7.8% of t values from samples of 10 fall beyond ± 2, against 4.9% from samples of 101, close to the 5% a normal curve gives. Small samples produce more large t values because s is sometimes well below σ, which shrinks the denominator. That is why a test with few degrees of freedom needs a larger cutoff (2.26 at $df = 9$) to keep false alarms at 5%.

Critical values

For two-tailed tests at $\alpha = 0.05$:

- z uses critical values ± 1.96.
- t uses wider cutoffs for small samples: ± 2.26 at $df = 9$, ± 2.09 at $df = 20$, ± 2.05 at $df = 29$. They approach ± 1.96 as n increases.

Rule of thumb. Under H_0, $|t|$ beyond about 2 happens about 5% of the time once df is 30 or more. With fewer df it happens more often, about 8% at $df = 9$.

t when H_0 is true

t can be positive or negative depending on direction. Under H_0, t follows the t distribution with $df = n-1$.

To see how t behaves when H_0 is true, we can simulate it. We set up a population whose mean really is $\mu_0 = 0$, draw a sample of 10, and compute one t. Then we repeat that 1,000 times.

Before you look. Each bar in Figure 2 counts t values from these samples. Decide roughly what share of them will fall beyond ± 2.26, the critical values for $df = 9$.

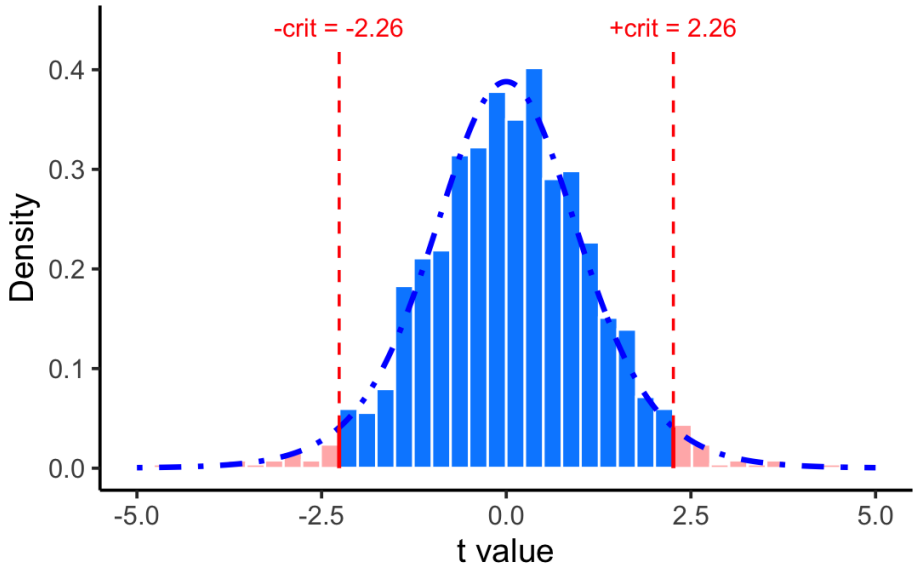


Figure 2: Histogram of 1,000 simulated t values from samples of 10 drawn from a population whose mean equals $\mu_0 = 0$. Bars beyond the critical values, ± 2.26, are shaded red. The blue dot-dash curve is the t distribution with 9 degrees of freedom.

Most of the 1,000 values fall between −2 and 2. About 4% fall beyond ±2.26, close to the 5% the cutoff is built to allow. The blue curve is the t distribution with 9 degrees of freedom, and the histogram follows it. Software uses that curve in place of a simulation to find how often chance alone produces a t as large as ours.

The one-sample *t*-test

Use a one-sample t when you want to test whether a sample mean differs from a reference value (μ_0). It asks whether the observed difference is bigger than you'd expect from sampling variability, given n and your sample SD s.

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \qquad df = n - 1$$

After calculating t from our data, we compare the observed t to a t distribution with $n-1$ degrees of freedom to get a p-value and a decision at α.

Calculating *t* from data

Let's calculate a one-sample t statistic step by step using a short example. A new highway opened near a residential area, and a noise guideline for the area sets a daytime level of 50 dB. Ten monitoring sites near the highway each recorded an average daytime sound level, in decibels (dB). The sound levels are invented. They are:

Sound levels (dB): 50, 70, 60, 40, 80, 30, 90, 60, 70, 60

- H_0: $\mu = 50$. The mean sound level near the highway equals the 50 dB guideline.
- H_a: $\mu \neq 50$.

By hand, the standard error of the mean is the sample SD divided by the square root of n:

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}} = \frac{17.92}{\sqrt{10}} = 5.67$$

And t is the difference between the sample mean (61 dB) and the reference value (the 50 dB guideline), divided by that standard error:

$$t = \frac{\bar{x} - \mu_0}{SE_{\bar{x}}} = \frac{61 - 50}{5.67} = 1.94$$

We can confirm this with a single R command:

```
t.test(sound_db, mu = 50)
```

Results: $t(9) = 1.94$, $p = 0.084$, mean = 61.0 dB, 95% CI [48.18, 73.82].

Confidence interval. The 95% CI in that output is $\bar{x} \pm t_{crit} \cdot SE_{\bar{x}}$, with t_{crit} taken from the t distribution with $n-1$ degrees of freedom. Here that is $61 \pm 2.262(5.67)$, or [48.18, 73.82]. The interval contains 50, so the test does not reject H_0. For a two-tailed test at α, the $100(1-\alpha)$% CI and the critical value always reach the same decision, the same equivalence Chapter 4 showed for z.

Interpretation. The mean (61 dB) is above the 50 dB guideline, but the difference is not statistically significant at $\alpha = 0.05$.

Assumptions check. The 10 sound levels spread evenly around 60 dB with no outliers, so approximate normality is reasonable. Section covers how to check this.

Interpreting *t*

Extreme sample means are unlikely if H_0 is true, which is why large $|t|$ values are rare. That is the basis of the decision rule for t-tests: if your observed t falls beyond the critical values, you reject H_0.

We're asking whether our t-value could plausibly have come from the **null distribution** of t, the distribution of t-values we'd see across many samples of size 10 if the true mean were exactly 50 dB, the guideline. It is the t version of Chapter 4's chance window and of the sampling distribution of the sample mean under H_0. It is a different curve from the null population distribution, which describes individual values. Figure 3 shows what that distribution looks like, at $df = 9$.

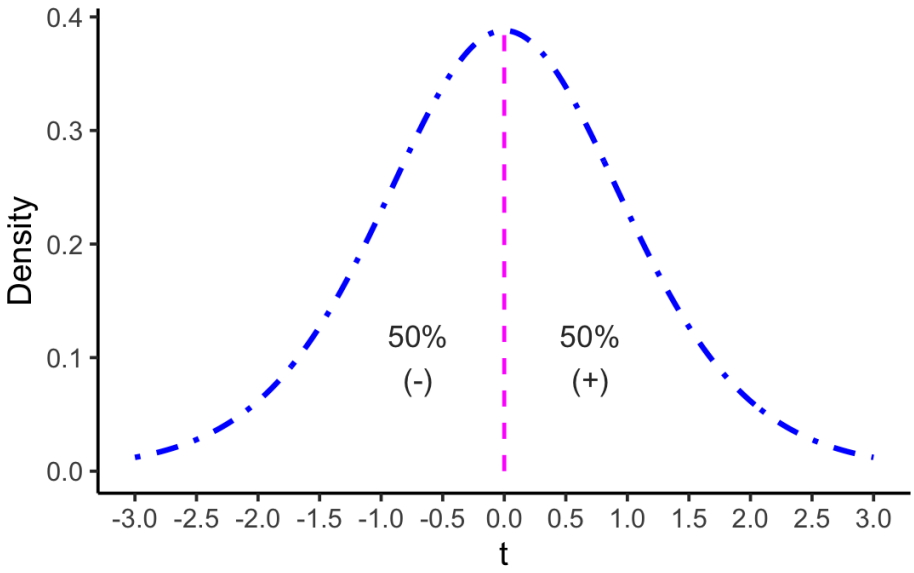


Figure 3: The distribution of t-values that chance alone produces when the population mean equals μ_0 ($df = 9$).

Under H_0, t is most likely to be near zero, because chance alone rarely produces a large difference. The distribution is symmetric, so half of all t-values are at or above zero and half are at or below.

We can get more precise than a 50/50 split. Figure 4 divides the same distribution into ten regions, each containing 10% of the distribution. Notice the regions get wider out in the tails: since most t-values cluster near 0, it takes a wider interval out there to capture the same 10%.

For example:

1. $t \leq -1.38$: $p = 10\%$
2. $-1.38 \leq t \leq -0.88$: $p = 10\%$
3. $-0.88 \leq t \leq -0.54$: $p = 10\%$
4. $t \geq 1.38$: $p = 10\%$

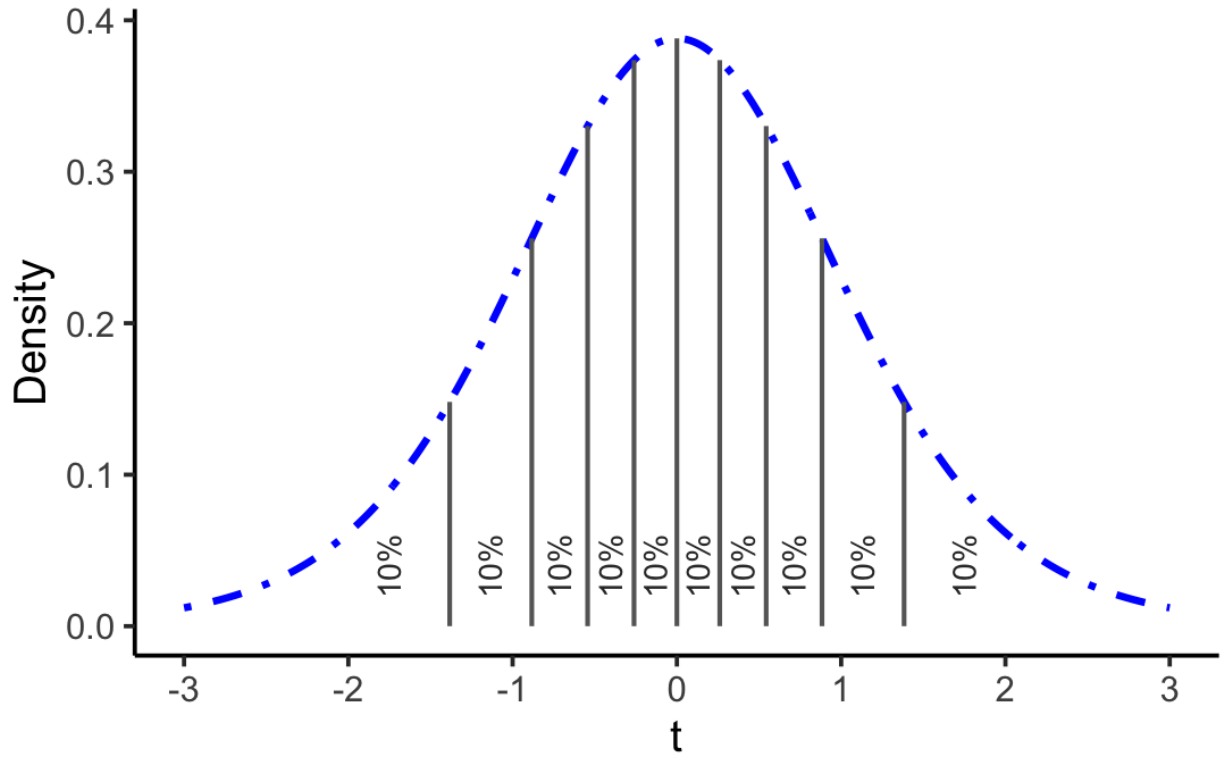


Figure 4: Splitting the t distribution up into regions each containing 10% of the t-values. The width between the bars narrows as they approach the center of the distribution, where there are more t-values.

Each range captures the same 10% of the distribution, even though the ranges themselves are different widths.

The highway noise t of 1.94 is beyond 1.38, so it falls in the top 10% of this distribution. Its exact p-value comes from software, and the one-tailed section below uses it.

Getting *p*-values for *t*

Older textbooks printed p-values for t in a table at the back of the book. Today R, Excel, and online calculators give them directly. You'll practice this in lab, both with software and by simulation.

Software needs two things from you to compute a p-value: the degrees of freedom, which you already have, and whether the test is one-tailed or two-tailed, which comes next.

One-tailed vs. two-tailed, applied

Chapters 4 and 5 covered when each is justified. For a t-test, the choice changes the critical value, and therefore the p-value, for the exact same data.

For the highway noise example ($df = 9$), a one-tailed test only needs to clear 5% of the distribution on one side. A two-tailed test has to clear 2.5% on each side, which pushes the critical value farther out:

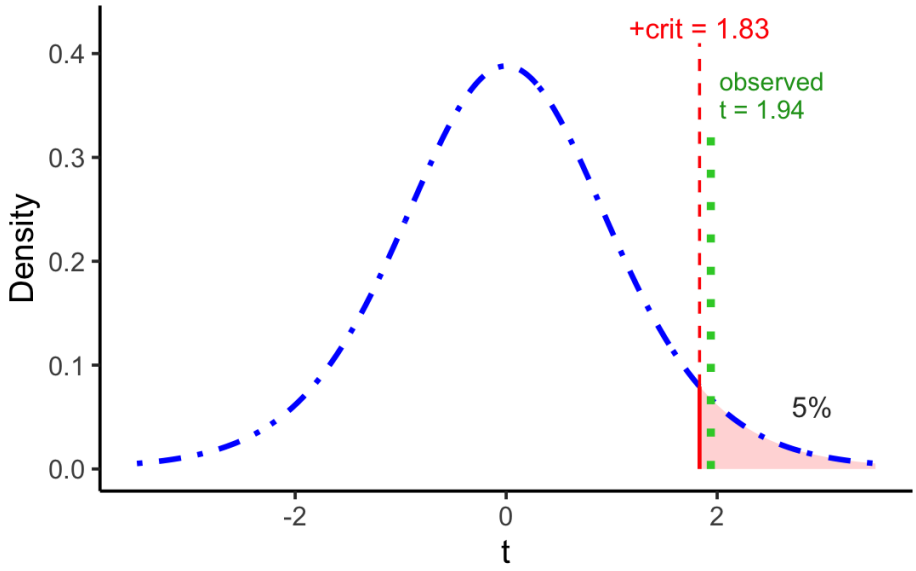


Figure 5: One-tailed critical value at $df = 9$. A t-value of 1.83 or greater occurs 5% of the time.

For our result, $t(9) = 1.94$, the one-tailed p is 0.042 and the two-tailed p is 0.084, exactly double, because a two-tailed test counts the matching region in both tails. The choice changes the decision here. One-tailed, the result is significant at $\alpha = 0.05$; two-tailed, it is not.

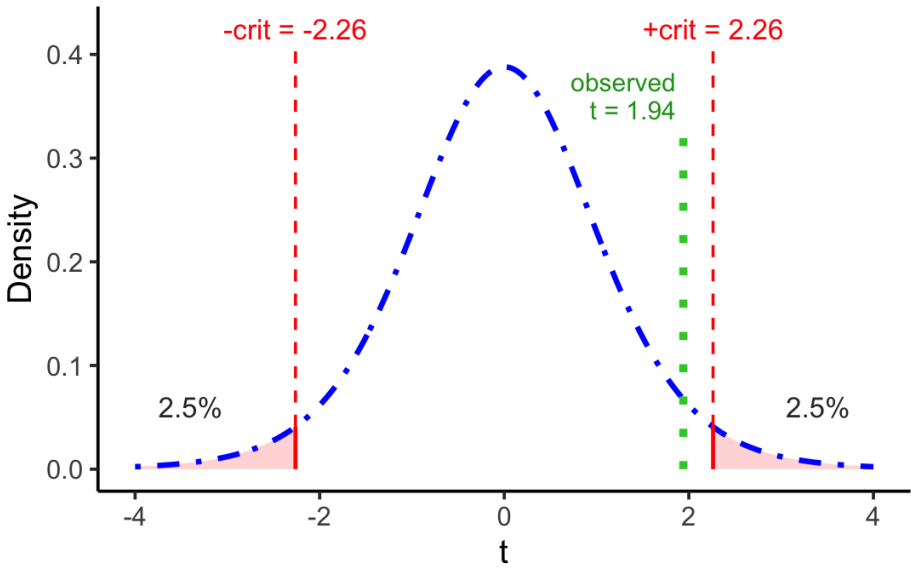


Figure 6: Two-tailed critical values at $df = 9$. Values of t beyond ± 2.26 occur 5% of the time in total, 2.5% in each tail.

Assumptions, and what to do when they don't hold

Every test in this book has assumptions. Assumptions are conditions that need to be roughly true for the math behind the test to be trustworthy. There are two different kinds of assumptions.

The universal assumption. Every test in this book, including the nonparametric tests in Chapter 7, assumes your observations are independent, so that knowing one observation's value tells you nothing about another's (Chapter 3's definition, applied to data), with the pairs as the observations in paired designs, and came from a sound study design (a fair sample, not one that systematically favors certain values). Because it applies everywhere, we won't list it test by test from here on. When an exam or homework question asks you to "list the assumptions of this test," it means the test-specific assumptions below.

Test-specific assumptions. These come from how a particular test's math works, and they differ test to test. For the one-sample and paired t-tests, the specific assumption is approximate normality of the outcome itself (one-sample) or of the difference scores (paired). The t-distribution we've been using to get p-values is *derived* assuming the population is normal, which makes the sample mean normally distributed too. If your data are badly skewed and your sample is small, that assumption fails, and the p-value may not be the true probability of a result this extreme under H_0.

How worried should you be? It depends on n. With a large sample, the sampling distribution of the sample mean tends toward normal even when individual observations are skewed (the central limit theorem), so the t-test is fairly **robust** to non-normality once n is reasonably large. Much environmental field data comes from small samples that are also skewed. Examples include a handful of expensive water samples, a short list of monitoring wells, or the five plots you had time to visit in a season. A t-test's answer becomes questionable when the sample is small and the data are also non-normal.

What to do about it. First, look at your data. A histogram or a Q-Q plot (Chapter 9 shows how to read one) will usually tell you whether "approximately normal" is a reasonable description or a stretch. You can also run a formal normality test, the **Shapiro-Wilk test** (`shapiro.test()` in R), which tests H_0: the data came from a normal distribution. Treat it as one more piece of evidence, though, rather than a strict gate. With a small sample, exactly when a violation matters most, it has low power and will often say "no problem" even when there is one. If your data look far from normal and your sample is small, you can

use a **nonparametric test** instead, which does not assume normality. You'll meet these nonparametric alternatives in the next chapter, once we've covered all three t-test forms below.

The paired *t*-test

When the same units are measured twice (before/after, matched pairs), the question is whether they changed. The natural data unit is the **difference score**, each unit's second measurement minus its first.

Design in this worked example. Ecologists measured soil moisture at 20 field plots before wetland restoration and again one year later, to see whether restoration increased moisture enough to support plant reestablishment and carbon sequestration. The soil moisture values are invented. Figure 7 shows the design.

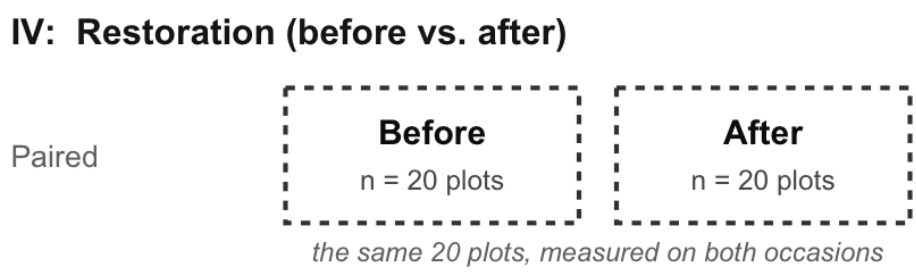


Figure 7: A paired design. The dashed boxes and the note underneath mark the feature that defines it: there is one set of 20 plots, measured on two occasions, not two separate samples.

The dashed boxes mark what makes this design paired. In every design so far the boxes held *different* units, and the test compared one set against another. Here both boxes hold the same 20 plots, so each plot's Before value can be subtracted from its own After value.

Define differences and their meaning. Choose an order and stick to it:

- $d_i = \text{After}_i - \text{Before}_i$
- $d_i > 0$: plot had more soil moisture after restoration (increase)
- $d_i < 0$: plot had less soil moisture after restoration (decrease)

This subtraction cancels between-plot variability (some plots are naturally wetter than others) and focuses on within-plot change. Because it removes those plot-to-plot differences, the paired test's standard error is smaller, so it has more power than an independent-samples test on the same data.

- H_0: $\mu_d = 0$. Restoration did not change soil moisture.
- H_a: $\mu_d \neq 0$.

Test on differences (paired t). Compute one difference per plot, then run a one-sample t against 0:

$$\bar{d} = \frac{1}{n}\sum_{i=1}^{n} d_i, \quad s_d = \text{SD}(d_1, \ldots, d_n), \quad t = \frac{\bar{d} - 0}{s_d/\sqrt{n}}, \quad df = n - 1.$$

The sign of t matches $\bar{d}$ and your difference direction.

What to report. $\bar{d}$ (the mean change), a 95% CI, $t(df)$, and p. Optionally add a standardized effect size, $\bar{d}/s_d$, the paired version of Cohen's d. The letter d does two jobs here. The d_i and $\bar{d}$ are difference scores, and Cohen's d is the Chapter 5 effect size.

Calculating *t* from the differences

To calculate t for a paired-samples t-test, we use the one-sample t-test formula from the previous section, applied to the difference scores instead of raw scores. The paired-samples t-test is a one-sample t-test on a single group of numbers, the differences.

Let's find t for the mean difference scores:

Table 1: Soil moisture (%) before and after restoration at five plots, with the steps for the paired t.

Plot	Before	After	Difference	Deviation from mean difference	Squared deviation
1	18	21	3	0.2	0.04
2	22	26	4	1.2	1.44
3	16	22	6	3.2	10.24
4	25	29	4	1.2	1.44
5	21	18	-3	-5.8	33.64
Sums	102	116	14	0	46.8
Means	20.4	23.2	2.8	0	9.36
				sd	3.421
				SE	1.53
				t	1.83

The five differences average 2.8 percentage points, with $s_d = 3.42$, so $SE_{\bar{d}} = 3.42/\sqrt{5} = 1.53$ and $t = 2.8/1.53 = 1.83$.

If we did this test using R, we would obtain almost the same numbers (there is a little bit of rounding in the table).

Table 2: Paired t-test on 5 restoration plots.

Mean difference	t	df	p
2.8	1.83	4	0.141

The mean difference was +2.8 percentage points, so soil moisture increased on average. Getting a p-value from this t, and choosing one-tailed vs. two-tailed, works exactly like it did for the one-sample case (see the one-sample t-test section above). Two-tailed, $t(4) = 1.83$, $p = 0.141$, which is not significant. With only five plots, a mean increase this large would not be unusual even if restoration had no effect on soil moisture.

Scaling to the full sample

The five plots above are a subset of the full restoration study, which measured all 20 paired plots. In R, give `t.test()` the two columns and set `paired = TRUE`. It subtracts the second column from the first, so this call tests After minus Before. Running `t.test(after_full - before_full, mu = 0)` on the differences gives the same result.

```r
t.test(after_full, before_full, paired = TRUE)
```

Table 3: Paired t-test on all 20 restoration plots.

Mean difference	t	df	p	95% CI lower	95% CI upper
1.9	2.5	19	0.022	0.31	3.49

At $n = 20$, $t(19) = 2.50$, $p = 0.022$, a significant result. The mean difference (1.90 percentage points) is smaller than the 5-plot subset's 2.8. The larger t comes from the standard error, which fell from 1.53 to 0.76 with four times as many plots. A mean increase of 1.9 points would be unlikely if restoration had no effect, so the extra plots gave more power to detect a real effect.

Confidence interval. The CI for a paired test follows the same recipe as the one-sample CI, the estimate plus or minus t_{crit} times the estimate's standard error. Here the estimate is $\bar{d}$, its standard error is $s_d/\sqrt{n}$, and t_{crit} comes from the t distribution with $df = 19$:

$$\bar{d} \pm t_{crit} \cdot \frac{s_d}{\sqrt{n}} = 1.90 \pm 2.09 \times 0.76 = [0.31, 3.49]$$

This is the interval R reports in the table above. It is the 95% CI for the effect, the mean difference in soil moisture, in percentage points.

The independent-samples *t*-test

The independent-samples t-test is the third and last design in this chapter. The logic is the same as the one-sample and paired cases: a mean difference divided by an estimate of its variability. Only the research design is different.

Use an independent-samples t-test for a **between-subjects** design: different people or units in each condition, each measured once. With two groups of 5, that's 10 units in total, none of them paired. Without repeated measures on the same unit, there's no meaningful way to subtract scores between conditions the way the paired t-test does.

The numerator is the mean difference between groups, as in the other two tests. The denominator differs from the paired case, because now we need a single estimate of variability built from two separate samples instead of one.

A natural first guess is to average the two groups' standard errors:

$$t = \frac{\bar{x}_A - \bar{x}_B}{\left(\frac{SE_A + SE_B}{2}\right)}$$

That average understates how much a difference between two means varies. The standard error of a difference between two independent means is $\sqrt{SE_A^2 + SE_B^2}$, which is always larger than the average of the two SEs. Instead of averaging, the independent-samples t-test pools the two groups' variances into a single estimate for the denominator. The pooled variance is Chapter 1's weighted mean of s_A^2 and s_B^2, with each weighted by its degrees of freedom:

$$t = \frac{\bar{x}_A - \bar{x}_B}{s_p \sqrt{\frac{1}{n_A} + \frac{1}{n_B}}}$$

where the pooled standard deviation s_p is (note s_A^2 and s_B^2 here are variances, not standard deviations):

$$s_p = \sqrt{\frac{(n_A - 1)s_A^2 + (n_B - 1)s_B^2}{n_A + n_B - 2}}$$

The pooled test has $df = n_A + n_B - 2$, one degree of freedom used for each group's mean. With two groups of 5, that is $df = 8$, the 8 in $t(8)$ below.

The pooled test adds two test-specific assumptions: the outcome is approximately normal within each group, and the two groups have similar variances. When the spreads clearly differ, use Welch's version instead.

The worked example compares bird species at ten field-edge sites. Five sites had their hedgerow removed (Group A), and five kept an intact hedgerow (Group B). At each site, one 10-minute point count recorded the number of bird species seen or heard. The counts are invented.

- H_0: $\mu_A = \mu_B$. Hedgerows make no difference to the mean number of bird species.
- H_a: $\mu_A \neq \mu_B$.

The code below follows the formula step by step in R, then checks it against `t.test()`. R reports the same t in the table below.

```
## By "hand" using R
# Bird species per point count
a <- c(1,2,3,4,5) # Group A, hedgerow removed
b <- c(3,5,4,7,9) # Group B, hedgerow intact
n_a <- length(a) # sample size for A
n_b <- length(b) # sample size for B

mean_difference <- mean(a)-mean(b) # compute mean difference

variance_a <- var(a) # compute variance for A
variance_b <- var(b) # compute variance for B

# Compute top part and bottom part of sp formula

sp_numerator <- (n_a - 1) * variance_a + (n_b - 1) * variance_b
sp_denominator <- n_a + n_b - 2
sp <- sqrt(sp_numerator/sp_denominator) # compute sp


# compute t following the formula

t <- mean_difference / (sp * sqrt(1/n_a + 1/n_b))
```

```
t.test(a, b, var.equal = TRUE)
```

Table 4: Independent-samples t-test (pooled) on bird species, Group A minus Group B.

Mean difference	t	df	p	95% CI lower	95% CI upper
-2.6	-2.02	8	0.078	-5.57	0.37

Our hand-computed t is -2.02, and R's t in the table matches it. R also reports $df = 8$, $p = 0.078$, and the 95% confidence interval for the difference.

Confidence interval. The recipe is the same as before, the estimate plus or minus t_{crit} times its standard error. The estimate is $\bar{x}_A - \bar{x}_B$, and its standard error, $SE_{\text{diff}} = s_p\sqrt{1/n_A + 1/n_B}$, is the denominator of the t just computed. With t_{crit} from the t distribution with $df = 8$:

$$(\bar{x}_A - \bar{x}_B) \pm t_{crit} \cdot SE_{\text{diff}} = -2.60 \pm 2.31 \times 1.29 = [-5.57, 0.37]$$

This is the interval R reports in the table above. It is the 95% CI for the effect, the mean difference in bird species between the two groups.

Cohen's d for two independent groups divides the mean difference by the pooled standard deviation, Cohen's $d = (\bar{x}_A - \bar{x}_B)/s_p$. Here $s_p = 2.04$, so Cohen's $d = -2.6/2.04 = -1.28$.

Table 5 sets the three forms side by side. Each one divides an effect by its standard error. The design decides what counts as the effect and how its standard error is estimated.

Table 5: The three t-tests compared.

	One-sample	Paired	Independent-samples
Design	One sample against a reference value	The same units measured twice	Two separate groups
Effect (numerator)	$\bar{x} - \mu_0$	$\bar{d}$	$\bar{x}_A - \bar{x}_B$
Standard error (denominator)	$s/\sqrt{n}$	$s_d/\sqrt{n}$	$s_p\sqrt{1/n_A + 1/n_B}$
df	$n - 1$	$n - 1$ (n = number of pairs)	$n_A + n_B - 2$
R call	`t.test(x, mu = 50)`	`t.test(after, before, paired = TRUE)`	`t.test(a, b, var.equal = TRUE)`

! Results reporting: t-tests

All three forms report the same things in the same order: the effect and its direction in plain language, the test statistic with its df, p, and the 95% CI for the effect. For the paired and independent-samples tests, that is the CI for the mean difference. The three forms differ only in what counts as the effect. One-sample, where the effect is the sample mean against the reference value:

> Mean sound level near the highway was 61.0 dB, which did not differ significantly from the 50 dB guideline, $t(9) = 1.94$, $p = 0.084$, 95% CI [48.18, 73.82] dB.

Paired, where the effect is the mean *difference* score:

> Soil moisture increased by 1.90 percentage points after restoration, $t(19) = 2.50$, $p = 0.022$, 95% CI [0.31, 3.49] percentage points.

Independent-samples, where the effect is the difference between two group means:

> Sites without a hedgerow had fewer bird species than sites with one, a mean difference of −2.60 species, $t(8) = -2.02$, $p = 0.078$, 95% CI [-5.57, 0.37] species, Cohen's $d = -1.28$.
>
> Note the paired write-up never mentions the before and after means separately. The difference score *is* the unit of analysis, so that single number plus its CI is the whole result.

Simulating data for t-tests

Earlier in this chapter we simulated t under H_0 and saw its null distribution in Figure 2. The same kind of simulation also gives a p-value for each repetition, and those p-values show what a test does when H_0 is true and when it is false.

Simulating under conditions you choose shows how sampling error affects results for a given sample size, expected difference, and amount of variation. Chapter 5 estimated power the same way and checked each estimate against the exact power computed from the t distribution.

Simulating a one-sample t-test

A simulation has two steps.

1. Decide what the population looks like. Here each sample is 30 sites, with values drawn from a normal distribution with mean 50 and SD 25.
2. Draw one sample from that population, run the t-test, and save the statistics you care about (t and p). Repeat 1,000 times and look at how they are distributed.

Every test compares the sample mean to $\mu_0 = 50$, which is also the true population mean, so H_0 is true in every repetition. Figure 8 shows the 1,000 p-values it produces.

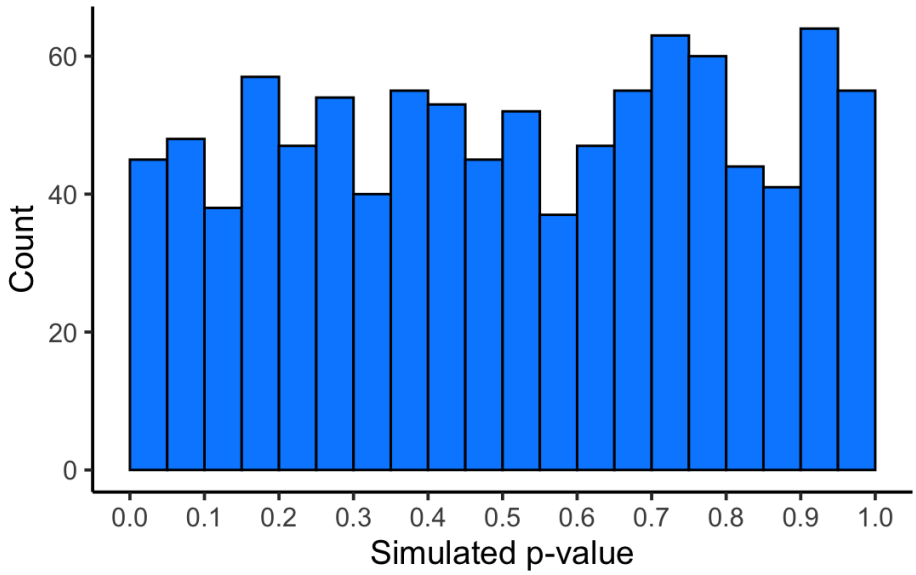


Figure 8: The distribution of p-values that are observed is flat under the null.

The p-values in Figure 8 are flat, as they should be when H_0 is true. This means you have the same chance of getting a t with a p-value between 0 and 0.05 as you would of getting one between 0.90 and 0.95. Those ranges are both ranges of 5%, so they hold an equal share of the t values by definition.

The first bar holds the 45 p-values below 0.05. Each one is a test that rejected a true H_0, a Type I error. Together they make up about 5% of the 1,000, which is the α we chose.

Simulating paired and independent-samples tests

The paired and independent-samples cases behave the same way, because they use the same t calculation on different inputs. A paired test runs a one-sample t on the difference scores; an independent-samples test swaps in the pooled standard error. Simulate either one under H_0 and you get the same two pictures: a t distribution centered on zero, and a flat p distribution.

Chapter 5 simulated 1,000 experiments with a real difference (two groups of 10, means 10 and 12.5, SD 5) and counted how often $p < 0.05$ to estimate power. Figure 9 shows the 1,000 p-values from the same setup.

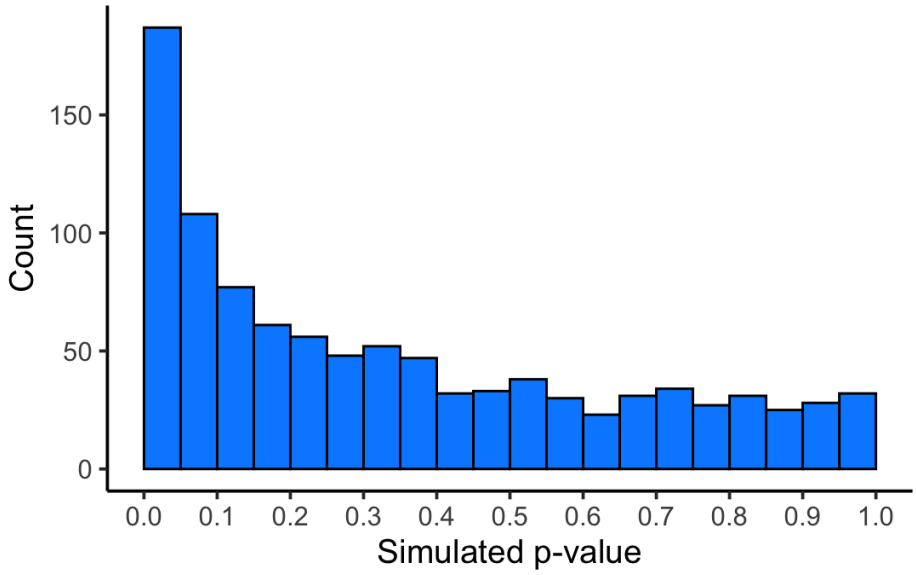


Figure 9: 1,000 simulated p-values ($n = 10$ per group) when there is a true difference.

With a real difference, the p-values pile up near 0, while under H_0 in Figure 8 they spread evenly from 0 to 1. Here H_0 is false, so rejecting it is a correct decision, not a Type I error. The first bar holds the 187 experiments that rejected H_0, and their share of the 1,000, 18.7%, estimates the design's power. The exact power for this design, computed from the t distribution, is 19%. The other 813 missed a difference that was really there. Chapter 5 showed how a larger sample raises power.

Chapter Summary

Why it matters. The t-test measures a signal against the noise around it. ANOVA, regression, and the rest of the course use the same logic.

Core ideas

- **Signal over noise.** A difference is only impressive relative to the variability around it. Every t statistic is a ratio: the effect you observed on top, an estimate of how much that effect could vary by chance on the bottom.
- **Three forms, one idea.** The one-sample, paired, and independent-samples t-tests differ in how they compute the effect and its standard error, but all three ask whether an observed difference is large compared to its own standard error. The paired test is just the one-sample test applied to difference scores.
- **Why t and not z.** When the population SD σ is unknown, we estimate it with s, which adds uncertainty. The t distribution accounts for that by being fatter in the tails than the normal, and it tightens toward the normal as degrees of freedom grow.
- **Assumptions come in two kinds.** *Independence and a sound study design* apply to every test in this book and are assumed from here on. The *test-specific* assumption is approximate normality: of the outcome (one-sample), of the difference scores (paired), or of each group (independent samples). The pooled independent-samples test also assumes similar variances.
- **How worried to be depends on n.** Thanks to the central limit theorem, t-tests are fairly robust to non-normality once samples are reasonably large. t-tests are least trustworthy when samples are small *and* skewed, which describes a lot of environmental field data. Plot your data first, and treat a Shapiro-Wilk result as evidence rather than a gate, since it has the least power with small samples, when a violation matters most.
- **The tail choice changes p for the same data.** For the highway noise example, $t(9) = 1.94$ gives a one-tailed p of 0.042 and a two-tailed p of 0.084. Decide which you need before seeing the result, using the rule from Chapters 4 and 5.

Student, A. 1908. "The Probable Error of a Mean." *Biometrika* 6: 1–2.

Welch, B. L. 1947. "The Generalization of 'Student's' Problem When Several Different Population Variances Are Involved." *Biometrika* 34: 28–35.